DR. NILESH SHAH - STOCKS FAIRSCOPE

US & Indian Stock Fair Value Desktop Application

Project report • Version 10.1 • 20 August 2026

Selected requirement	Implemented solution
Markets	US equities, NSE, and BSE
Fair-value strategy	Sector-aware blend of two-stage DCF/residual income, Graham/book, and live peers
Assumption mode	Automatic base assumptions plus conservative/base/optimistic scenarios
Data source	Yahoo primary plus Screener.in and NSE/BSE website verification; no API key
Currency & uncertainty	Native currency; dynamic 15–50% risk/data-aware valuation buffer
Delivery	Modern Qt app plus Windows x64 signing-ready Inno Setup pipeline
Reports	Full per-stock PDF export; included live AAPL sample
Batch analysis	Import up to 100 CSV/TXT/Excel symbols; export CSV and branded PDF summary
Search	Debounced company/ticker autocomplete for US, NSE and BSE
Visual analysis	Native Qt and PDF charts for price, models, trends, sensitivity and Monte Carlo
Assumption review	Automatic-vs-reviewed table, presets, locks, validation, notes and PDF disclosure
Data lineage	Metric-level source URLs, periods, transformations, statuses, CSV and PDF citations
Model audit	Unique IDs, parent chain, versions, events, seed, JSON export and SHA-256 verification
Financial companies	Specialized bank/NBFC/insurer data, valuation, risk, lineage, audit and PDF review
Coverage fallback	Revenue, book/net-cash and peer P/B heuristics for loss-making/negative-FCF equities
External review	User-cited Moneycontrol/filing values with lineage, child audit ID and no automated scraping
Watchlists	Local SQLite watchlist with state-aware valuation-buffer, price, upside, risk, discrepancy and FV-change al
Corporate monitor	SEC/NSE/BSE/Screener filings, announcements, dividends, splits and persistent read history
Local database	WAL SQLite stock cache, stale fallback, settings, and 1,000-record valuation audit history
Background jobs	Bounded parallelism, retry/backoff, determinate progress, pause/cancel, partial persistence and recovery
Testing	Cross-platform Python 3.11–3.13 CI, coverage, PDF/import/extreme/cache/100-stock/network regression s
Diagnostics	Sanitized rotating logs, provider/database health, maintenance, support ZIP and colourful watchlist PDF
Privacy	OS-keychain/Fernet secrets, restrictive permissions, masked watchlist UI, redacted support exports
PDF design	Branded cover, executive summary, contents, bookmarks, score bars, charts, audit and lineage appendice
Uncertainty language	Valuation-buffer terminology, no Buy/Sell labels, and explicit no-guarantee/loss-protection warnings
Phase 2	Sensitivity matrix, reverse DCF, business quality, governance screen, numeric confidence

Selected requirement	Implemented solution
Simulation	2,000-run deterministic Monte Carlo valuation with percentile ranges

Product overview

DR. NILESH SHAH - STOCKS FAIRSCOPE is a responsive desktop research tool designed to turn a ticker into a transparent scenario valuation. The modern Qt interface includes visual analysis, lineage/audit review, and specialized bank, NBFC, lender, and insurer workflows. Yahoo Finance requests and batch work use Qt thread-pool workers so the window remains responsive.

Primary workflow

• Choose US, NSE, or BSE and enter a ticker.

• Retrieve latest price, profile, market metrics, cash flow, EPS, book value, and growth indicators.

• Apply market-aware guardrails and calculate three scenarios.

• Review model-level and blended values, upside/downside, and valuation-buffer threshold.

• Optionally edit assumptions and regenerate the analysis, then export a full PDF.

Phase-1 valuation design

Normalized fundamentals

Up to five annual free-cash-flow, revenue, and net-income observations are collected. Median normalization reduces one-year distortions, while historical and reported growth are combined under sector-specific caps.

Two-stage intrinsic value

Operating companies use a ten-year normalized FCF-per-share DCF: initial growth applies for five years and then fades toward terminal growth. Financial institutions use residual income instead.

Live peer comparison

Real peers use Yahoo screening. Indian prices and selected ratios are cross-checked against Screener.in and NSE/BSE website endpoints with discrepancy warnings. Median P/E is used for most companies; financial and real-estate profiles use median P/B.

Sector and uncertainty profiles

Financial, real-estate, cyclical, utility, technology, and standard-company profiles use different weights and limits. The required valuation buffer increases for leverage, weak coverage, FCF instability, incomplete data, cyclicality, and model disagreement; it does not guarantee loss protection.

Phase-2 analytics

A 5×5 valuation sensitivity matrix, reverse DCF implied growth, business-quality score, governance-risk screen, and expanded numeric confidence score are shown in the app and reports.

Monte Carlo uncertainty analysis

Every stock receives a 2,000-run reproducible Monte Carlo valuation distribution with percentile ranges. These are model outputs, not return forecasts.

Scenario rules

Scenario	Growth	Discount rate	Terminal growth
Conservative	Base – 4 percentage points	Base + 2 points	Base – 1 point
Base	Bounded data estimate	10% US / 12% India	3% US / 5.5% India
Optimistic	Base + 4 points	Base – 1.5 points	Base + 1.5 points (guarded)

Resilience

When a model lacks required positive inputs, it is excluded and remaining weights are re-normalized. The app reports model coverage and data warnings rather than silently replacing missing fundamentals. Input guardrails require discount rate above terminal growth, a 1–20 year forecast horizon, and valid non-negative weights.

Delivery, testing & limitations

Included files

The downloadable ZIP contains source, Windows x64 build/sign/verify scripts, dependency list, Windows and Unix launchers, single-stock and batch PDF exporters, CSV/TXT/Excel import support, offline unit tests, setup guide, this report, and a sample AAPL valuation PDF generated from the live feed.

Verification completed

• Offline tests cover scenarios, dynamic valuation buffer, peers, sector selection, reverse DCF, sensitivity, quality/governance, Monte Carlo, batch import, and missing-model reweighting.

• All Python modules passed byte-code compilation.

• Yahoo lookups were verified for US/NSE/BSE; Screener.in and BSE website checks were validated, while NSE access remains location-dependent.

• The full PDF report generator was executed successfully for AAPL.

Known limitations

• Free Yahoo/Screener/NSE/BSE website data can be delayed, unavailable, incomplete, rate-limited, blocked, or changed without notice.

• A bounded P/E normalization is not the same as a researched peer-set median.

• Banks, insurers, REITs, cyclicals, loss-making firms, and unusual capital structures often need specialized valuation methods.

• Intrinsic value is assumption-sensitive and is not a forecast or guarantee.

• An executable must be built on the target operating system; source launchers are supplied for portability.

Disclaimer

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